

Faculty of Science, Engineering and Built Environment

SEE314 Energy Efficiency, Management and Market Analysis

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

SEE314 provides students with an overview of energy efficiency, demand side management systems as well as energy market, while incorporating the strategies to develop sustainable energy systems with increasing penetration of renewable energy. Students will be required to complete energy efficiency calculations, analyse data, and apply their understanding of energy management to make recommendations to stakeholders. They will also design appropriate demand response strategies which will be essential for effective operation of electricity networks integrated with renewable energy. Students will learn about the operations of energy markets along with government policies for energy systems. They will have the opportunity to investigate the impact of transmission network on energy market, economic operation of power systems, and the principles of ancillary service markets. The cases studied will include real-world examples from the Australian Energy Market Operator (AEMO) and renewable energy microgrid projects. By solving recent challenges in energy industries associated with market reforms and policies, students will develop critical thinking and analytical abilities to recommend new strategies for improving energy efficiency and energy market operation.

This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Saman Asghari Gorji
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Phone: +61 3 522 72383
Other members of the team and how to contact them

Other members of the teaching team will be announced on the unit site.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link). Student evaluation surveys are conducted at Deakin to assess students' satisfaction with their teaching and learning experiences and to assist in determining whether any improvements are required. This unit will be offered for the first time in 2025 therefore no previous student evaluations have been completed. I encourage you to complete the survey as the

results are used to help assess curriculum, teaching and assessment directions in this unit.
If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Design solutions for energy efficiency, demand management and energy markets and recommend associated reform policies.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking GLO5: Problem solving
ULO2	Perform sensitivity analysis using industry standard software to improve the energy efficiency of residential, commercial or industrial buildings.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy GLO4: Critical thinking
ULO3	Develop and analyse business models for energy markets.	GLO4: Critical thinking
ULO4	Collaboratively develop Innovative strategies for improving energy efficiency, demand management, as well as energy market operations and communicate recommendations through a technical report.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO4: Critical thinking

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit students must achieve a minimum of 50% in the design problem solving test.

Brief summary of the hurdle requirements	Rationale
To be eligible to obtain a pass in this unit students must achieve a minimum of 50% in the design problem solving test.	This unit requires a pass in the problem solving tasks to ensure that each student can demonstrate their learning independently.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Written report based on Software-based Design activities**
Brief description of assessment task	Students will submit a written report based on software-driven design activities recommending energy efficacy strategies and applying MATLAB for energy market or policy analysis.
Student output	Individual written report, incorporating sensitivity analysis and simulation results

Grading and weighting (% total mark for unit)	30% (Graded)
This task assesses your achievement of these Unit Learning Outcome(s)	This assessment evaluates students' ability to conduct sensitivity analysis using industry-standard software to enhance energy efficiency in different building types (ULO2). Through the application of MATLAB, students will analyse energy performance, optimise consumption, and recommend efficiency improvements.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Through student ability to apply discipline-specific knowledge and capabilities on energy efficiency and energy market policy analysis GLO3: Through student ability to perform software simulation for analysing energy efficiency and energy market scenarios GLO4: Through student ability to critically reflect on the analysis outcomes of energy efficiency and energy market policies
How and when you will receive feedback on your work	Ongoing feedback during seminars and lab activities, with final written feedback within two weeks of submission.
When and how to submit your work	Due by **8pm AEST,** Friday April 11, 2025 **Week 6** To be submitted online via the dropbox in unit site

Summative assessment task 2

Title	**Problem Solving Test**
Brief description of assessment task	Students will solve a set of numerical and analytical questions related to energy efficiency, energy markets and demand management, focusing on reform policies.
Student output	Written response, including conceptual analysis and numerical calculations. Students will have 2 hours to complete each test. The test will be delivered online via the unit site.
Grading and weighting (% total mark for unit)	40% (Graded)
This task assesses your achievement of these Unit Learning Outcome(s)	The problem-solving test assesses students' ability to design solutions for energy efficiency, demand management, and energy markets while recommending reform policies (ULO1). It also evaluates their capacity to develop and analyse business models for energy markets (ULO3).
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Through student ability to apply discipline-specific knowledge and capabilities by designing solutions for energy efficiency problems and recommending effective energy market policies GLO4: Through student ability to apply critical thinking skills to compare different solutions and identify best possible strategies to improve energy efficiency and address the energy market challenges GLO5: Through student ability to propose solutions for real-world energy efficiency and energy market problems
How and when you will receive feedback on your work	Feedback will be provided after submission and discussed during seminars.
When and how to submit your work	Problem solving test will be scheduled during **Week 9.** Further details will be provided on the unit site.

Summative assessment task 3

Title	**Team Case Study Report**
Brief description of assessment task	Students will work in groups to submit a written report based on a real-world case study. The focus will be on energy efficiency strategies within energy markets.

Student output	Team written report (approx. 4,000 words), including analysis of energy efficiency and energy market strategies as well as energy management.
Grading and weighting (% total mark for unit)	30% (Graded)
This task assesses your achievement of these Unit Learning Outcome(s)	This group-based assessment requires students to collaboratively develop innovative strategies for energy efficiency, demand management, and energy market operations (ULO4). It also involves performing sensitivity analysis (ULO2) and analyzing business models for energy markets (ULO3) within a real-world case study.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Through student ability to apply discipline-specific knowledge and capabilities for a real-world energy efficiency and energy market case study analysis GLO2: Through student ability to communicate key findings from investigation of the energy efficiency and energy market case study GLO3: Through student ability to apply software based design tools for creating effective solutions in the energy efficiency and energy market case study GLO4: Through student ability to reflect critically on the justification of the proposed solutions for energy efficiency and energy market case study.
How and when you will receive feedback on your work	Feedback will be provided progressively during seminar sessions and within two weeks of submission.
When and how to submit your work	Due by **8pm AEST,** Friday May 30, 2025 **Week 12** To be submitted online via the dropbox in unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 1 hour seminar per week, 4 x 3 hour practical experiences (laboratories) per trimester (1 x 3 on campus and 3 x 3 online)
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 1 hour seminar per week, 4 x 3 hour online practical experience (laboratory) per trimester.
Typical study commitment

Students will on average spend 150 hours over the trimester undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.

Students engage in a mix of theoretical and practical learning. Seminars (on-campus) provide structured discussions and problem-solving exercises, while practical sessions focus on real-world energy efficiency and market simulations. Students attend one in-person session (Week 5), while the remaining sessions are conducted online (Weeks 9, 10, and 11). Assessment tasks reinforce learning through individual reports, problem-solving exercises, and a group case study, ensuring students develop analytical skills, apply industry software tools, and propose innovative solutions for energy market challenges.
Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SEE314 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

Software Tools: MATLAB – Required for sensitivity analysis, energy efficiency modelling. (Available via Deakin Software Library).

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.
Suitable reference books for this unit are available in the University Library unit reading list and include:

- Wayne C. Turner and Steve Doty, "Energy Management Handbook", 9th Edition, CRC Press, 2018, ISBN 9781138666979
- Clark W. Gellings, The Smart Grid: Enabling Energy Efficiency and Demand Response, 1st Edition, CRC Press, 2009, ISBN 1439815747
- Arturo Losi, Pierluigi Mancarella, Antonio Vicino, Integration of Demand Response into the Electricity Chain, 1st Edition, Wiley, 2015, ISBN 9781848218543
- Fundamentals of power system economics, Book by Daniel Sadi Kirschen; Goran Strbac 2018, 2nd ed.
- Energy Management and Efficiency for the Process Industries – Alan P. Rossiter & Beth P. Jones
- Electricity Markets and Power System Economics – Deqiang Gan, Donghan Feng, and Jun Xie

Class and Seminar Notes: Weekly lecture slides and supplementary notes will be available on the unit site.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025

Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Special learning activities	Assessment due date
1	3 March 2025	Fundamentals of Energy Efficiency	- **Class:** Energy Efficiency Principles & Calculation. - **Seminar:** Problem-solving on system and component levels	
2#	10 March 2025	Energy Management Strategies Part I (Without Renewables)	- **Class:** Demand-Side Management & Demand-Response Part I. - **Seminar:** Problem-solving on demand response I	
3	17 March 2025	Energy Management Strategies Part II (With Renewables)	- **Class:** Demand-Side Management & Demand-Response Part II. - **Seminar:** Problem-solving on demand response II	
4	24 March 2025	Energy Efficiency & Conservation in Buildings	- **Class:** Smart Building & Energy Audit. - **Seminar:** Problem-solving on smart energy systems	
5	31 March 2025	Sensitivity Analysis & Energy Performance	- **Class:** Energy Consumption & Sensitivity Analysis. - **Seminar:** Problem-solving on sensitivity analysis - **Studio:** Sensitivity Analysis with MATLAB (face-to-face).	
6	7 April 2025	Fundamentals of Economics and Energy Market	- **Class:** Introduction to Economic Theories in Energy Markets. - **Seminar:** Problem-solving on market principles.	**Assessment 1: Written report** due 11 April
7*~	14 April 2025	Organisation of the Electricity Market	- **Class:** Structure & Operation of Electricity Markets. - **Seminar:** Problem-solving on electricity market structure.	
8	28 April 2025	Participation in the Electricity Market	- **Class:** Market Mechanism. - **Seminar:** Problem-solving	
9	5 May 2025	Transmission Lines & Electricity Market	- **Class:** Transmission Networks - **Seminar:** Revision for Assessment 2. - **Studio:** Energy Markets I.	**Assessment 2: Problem solving test** further details provided on the unit site

Week	Commencing	Topic	Special learning activities	Assessment due date
10	12 May 2025	Investment in Electricity Markets	- **Class:** Investment Strategies - **Seminar:** Energy Export II - **Studio :** Energy Markets II.	
11	19 May 2025	Frequency Control Ancillary Services Market	- **Class:** Managing Power System Frequency & Ancillary Services. - **Seminar:** Energy Export II - **Studio:** Energy Markets III.	
12	26 May 2025			Study period **Assessment 3: Team Case study report** due 30 May
13-14^	2 June 2025			End-of-unit assessment and exam period

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed